

Supplementary material for

Scalable studentized bootstrap variance inference under skewness and heavy tails

Additional computational benchmarks

This supplement reports additional runtime benchmarks that complement the main article’s complexity analysis (Section 4) and computational results (Section 5). All timings were obtained in R 4.4.1 on a 16-core AMD EPYC 7543P server with 128 GB RAM using parallel execution via `doParallel`/`doRNG`.

Table 1: Comparison of computational complexity and scalability: status quo vs. proposed SPDV approach for studentized bootstrap variance inference.

Method	Complexity	$n = 10^5, B = 10^4$	$n = 10^6, B = 5 \times 10^4$	Practical status
Direct pairwise evaluation	$\mathcal{O}(n^2)$	$\approx 19.5 \text{ min}^\dagger$	$\approx 162.5 \text{ h}^\dagger$	Impractical for big data
SPDV (proposed)	$\mathcal{O}(n)$	78.6 s	1.3 h	Routine big data inference

Timings for $n = 10^5$ are based on measured benchmarks using the SPDV- s^2 identity on standard parallel hardware (16-core CPU). Extrapolations for $n = 10^6$ assume quadratic scaling for direct pairwise evaluation and linear scaling for SPDV.

[†] Extrapolated from an $n = 10^4$ baseline.

Table 2: Runtime comparison (wall-clock seconds) for $B = 10,000$ studentized bootstrap replications.

Sample size n	Direct pairwise evaluation $\mathcal{O}(Bn^2)$	SPDV linear $\mathcal{O}(Bn)$
1 000	52.4	1.8
5 000	>600	8.9
10 000	infeasible	18.2

Direct evaluation of the pairwise variance representation becomes impractical beyond $n \approx 5,000$ due to quadratic scaling. The linear-time SPDV approach remains feasible for much larger n , consistent with the main article’s benchmarks.

Together, these benchmarks illustrate that direct pairwise evaluation rapidly becomes infeasible at modern data scales, whereas the SPDV implementation maintains practically acceptable runtimes even for very large (n, B) combinations.

Table 3: Wall-clock times for full Monte Carlo scenarios ($M = 5,000$, $B = 5,000$) using the SPDV $\mathcal{O}(n)$ implementation.

Sample size n	Serial (s)	Parallel 16-core (s)	Speedup	Direct $\mathcal{O}(n^2)$ est. (h)
20	0.12	0.08	$1.5\times$	< 0.01
50	0.31	0.15	$2.1\times$	< 0.01
100	0.62	0.22	$2.8\times$	0.02
10,000	62.1	4.2	$14.8\times$	173
100,000	621	39	$15.9\times$	173,000

Direct $\mathcal{O}(n^2)$ times are extrapolated from an $n = 100$ baseline under quadratic scaling. Serial times are measured without a parallel backend; parallel times use `doParallel/doRNG` on 16 cores.